

CS 280
Programming Assignment 2

Implementing the Backpropagation Algorithm

Instructor: Pros Naval
Due: October 18, 2016

In this Programming Assignment, you will implement an Artificial Neural Network, train it using the Backpropagation Algorithm on a given dataset and compare its performance on a classifier of your choice.

Instructions

1. Unzip the file CS280_PA3.zip that was provided to you.
2. Study the code named `cs280_neural_net.m`. Note that this is only a demo code meant to illustrate a simple neural network. You should be guided by the notes you took during our lecture on ANN.
3. Rewrite from scratch the code in C, C++, Python or R and revise so that it can accept numeric inputs. You are not allowed to use any code available anywhere (internet, book, friend, etc).
4. Use a spreadsheet to view the contents of the following dataset files:

data.csv : 3486 instances each having 354 attributes or features
data_labels.csv : class labels (1,2,...8) for each of the 3486 instances
test_set.csv : test instances without labels

5. Partition the dataset *data.csv* into a training set named *training_set.csv* and *training_labels.csv* and a validation set named *validation_set.csv* and *validation_labels.csv*. Please note that we have here a highly imbalanced dataset (i.e., there are huge differences in the number of instances for each class). Ideas on how to deal with this are found in <http://stats.stackexchange.com/questions/109177/which-classifiers-work-well-with-unbalanced-data?rq=1>
6. Train your ANN on the training set and use the validation set to monitor your prediction errors over training epochs.
7. Change the number of nodes in each hidden layer and the learning rate parameter so that the prediction error over the validation set is minimized. Once the network has learned, predict the labels for each of the test set instances. Save these labels in a csv file named *predicted_ann.csv* with the same format as *data_labels.csv*.
8. Choose another classifier (SVM, Random Forest, etc.) that you think will most accurately predict the correct class labels for the problem at hand. You may download code (Python, C, etc) or use a package (R), toolbox (Matlab/Octave) for your chosen classifier.

9. Train your selected classifier and predict the labels for the test set. Save the predicted labels in a csv file named *predicted_other.csv*.

Deliverables:

- a) ANN code (in C, C++, Python or R) with self-documenting comments. Please indicate compiler and OS you used.
- b) The following csv files:
 - i) *training_set.csv* and *training_labels.csv*
 - ii) *validation_set.csv* and *validation_labels.csv*
 - iii) *predicted_ann.csv* and *predicted_other.csv*
- c) Documentation (in the same style as the one for PA1) describing the following:
 - i) how the dataset was partitioned
 - ii) (ANN): how the number of nodes in the hidden layers was selected
 - iii) (ANN): plot of training and validation errors vs epoch number
 - iv) how you dealt with the highly imbalanced dataset
 - v) justification for your choice of the second classifier and details on how it was trained
 - vi) tables comparing the two classifiers in terms of their accuracies and running times
 - vii) Conclusion

Note: Make sure that your ANN code in (a) works correctly, otherwise the rest of your submission will not be checked. Submit the deliverables to submit2pcnaval@gmail.com with “[CS280 PA3 Submission] <Your_Name>” on the subject line by 11:59 pm of October 18, 2016.